

Paper 3

- 1 (a)** The amount of thermal energy required to convert the substance from liquid to vapour phase without a change in temperature.

Note: Candidates should be aware of difference between latent heat and specific latent heat.

- (b) (i)** This is to allow power loss to be eliminated from the calculation of the specific latent heat of the vaporisation. Since the temperature in both instances is the same, the power loss is also the same.

- (ii)** Heat absorbed by the boiling water

$$Q = Pt = ml_v$$

$$(140 - P_{loss})(300) = (14.1 \times 10^{-3})l_v$$

$$(95 - P_{loss})(300) = (8.2 \times 10^{-3})l_v$$

$$l_v = 2.3 \times 10^6 \text{ Jkg}^{-1} = 2300 \text{ Jg}^{-1}$$

Note: Need to be aware of basic procedures for determination of latent heat.

- 2 (a) (i)** The tension in the rod alone is not sufficient to provide the centripetal force